

Internet Programming (Sessional) [CSE-326]



Proposal Report on:
SmartPrint

Group No: 4

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Objectives

The primary objective of our web application, **SmartPrint**, is to create a user-friendly, secure, and scalable platform where people can seamlessly print anything from anywhere with just a tip of command. The web platform is designed to:

- Design and implement SmartPrint, a web-based front end that lets users submit print jobs from anywhere to a networked printer.
- Integrate the website seamlessly with CUPS running on Raspberry Pi OS so that jobs drop straight into the Pi's print queue.
- Automate job management—authentication, queueing, status updates, error-handling and billing (if needed)—to minimize manual intervention.
- Ensure the system is secure, reliable, and scalable, so additional printers or Pi hosts can be added with minimal configuration.

Introduction

SmartPrint is a fully automated printing management system built around a Raspberry Pi running CUPS (Common UNIX Printing System). Users visit a responsive website, authenticate, upload their documents (PDF, DOCX, images, etc.), choose print options (color/BW, duplex, paper size), and submit. Behind the scenes, the web server communicates with CUPS on the Pi, spooling jobs immediately to the attached printer. Real-time status (queued, printing, completed, error) is shown to users, and administrators can monitor and manage all jobs centrally. By offloading the print logic to CUPS and keeping the front end light, SmartPrint delivers a swift, robust solution that requires no client-side drivers—just a browser.

Motivation

SmartPrint aims to solve all of these challenges:

- Inefficient workflows: Currently, users either copy files to USB drives or install printer-specific drivers on every device—time-consuming and error-prone.
- Lack of remote access: In many small labs, the only way to print is to be physically on-site or jump through complicated VPN setups.
- High maintenance: Administrators must manually clear stuck jobs, install drivers, and help troubleshoot network-printer connectivity.
- Cost pressures: Small offices, libraries, and campus labs need a low-cost, open-source alternative to commercial “print management” suites.

Our motivation is to make everyday people's life easier, more organized, and more connected through SmartPrint.

Potential Stakeholders

The following groups are considered key stakeholders for our web application:

- End Users: Students, faculty, staff, or employees who need quick, remote printing without installing drivers.
ents can sell and purchase t-shirts and other products in a timely manner.
- IT Administrators: Campus or office IT teams who must manage printers centrally with minimal overhead.
- Library & Lab Supervisors: Who oversee shared printers and want usage logs, quotas, or billing data.
- Small Businesses & Startups: Looking for an affordable, maintenance-free printing solution.
- Event Organizers: That offer on-demand printing (e.g., conference badges) via kiosks powered by a Pi.

Existing Related Work

Currently, People mostly rely on manual printing where there is person to command the printer to print. These methods are unorganized, not searchable, and sometimes inaccessible to the wider student body.

Our platform addresses this issue by offering a centralized web application that includes:

- Google Cloud Print (deprecated): Once offered driverless cloud-based printing but was shut down in 2020, leaving a gap for lightweight replacements.
- PaperCut MF / NG: Commercial print management with quotas, secure-print release, and reporting—but costly and heavy for small deployments.
- Mopria Alliance & IPP Everywhere: Standards enabling driverless printing, but require compatible printers or intermediary software.
- OpenPrint: A community project to wrap CUPS in a modern web UI, but lacks polished authentication, mobile support, and job-tracking features.

Contribution of the Work

This project introduces a multifunctional web platform that:

- Pi-Powered Appliance: A turnkey system that runs entirely on a \$35 Raspberry Pi—no additional servers needed.
- Seamless CUPS Integration: Exposes every CUPS option through an intuitive UI, while handling edge cases (paper jams, low-ink) automatically.

- **Driverless, Anywhere Access:** Users only need a browser and credentials—no OS-specific drivers or VPN.
- **Extensible Architecture:** Modular codebase allowing future plugins (e.g., cost-per-page billing, mobile-app support, secure print-release via PIN).
- **Open Source:** All code, deployment scripts, and configuration are released under an MIT-style license, encouraging community contributions.

Proposed Features of the Web Application

Our web application, **SmartPrint**, is designed with multiple interconnected features aimed at providing a seamless and enriching user experience for the CUET community. These features focus on two main aspects: a digital.. Below are the key proposed features of the platform:

- **User Authentication & Roles**

University/office SSO or local account management

Roles: User, Lab-Manager, Administrator

- **Document Upload & Print Options**

Support for PDF, DOCX, JPEG, PNG, plain text

Options: color vs. B/W, single- vs. double-sided, paper size, number of copies

- **Real-Time Job Monitoring**

Queue dashboard showing status (Queued, Printing, Completed, Error)

Automatic email or SMS notifications on completion/failure

- **Error Handling & Recovery**

Detect paper jams, offline printer, low-ink; notify admins and retry automatically.

- **Usage Tracking & Reporting**

Logs of every job: user, time, pages, settings

Exportable reports (CSV/PDF) for billing or auditing

- **Admin Panel**

Add/remove printers and Pi hosts

Manage user quotas and prints-per-user limits

Clear or reprioritize jobs

- **Mobile-Responsive Interface**

Responsive design for smartphones and tablets

Optional dedicated mobile-app integration via REST API

- **Scalability & High Availability**

Ability to federate multiple Pi appliances under a single web UI

Fail-over support if one Pi goes down

- **Security & Privacy**

HTTPS everywhere with Let's Encrypt

Secure file cleanup after printing

Audit trails for compliance

- **Installer & Configuration Scripts**

One-click setup scripts to install Raspbian OS, CUPS, web server, SSL certs, and SmartPrint code.

Expected UI Design

Sign in and Sign Up Pages:

SmartPrint Home Login Sign Up

Welcome Back
Sign in to continue to SmartPrint

Email address

Password

☐ Remember me

Sign in

Don't have an account? [Sign up now](#)

SmartPrint
Smart printing solution for students and professionals.
f t @ in

Quick Links
Home
Dashboard
Printers
Print Jobs

Help & Support
FAQ
Contact Us
Privacy Policy
Terms of Service

Account
My Profile
Settings
Login

© 2025 SmartPrint All rights reserved. Made with ❤️ by SmartPrint Team

Upload Your File:

SmartPrint

Printers

Upload

Calculator

Login

Sign Up

Upload Your Files

Drag & drop PDF files here or [browse](#)

Only PDF files allowed

Uploaded Files

Total Pages: 0

Print Command:

SmartPrint

Printers

Upload

Calculator

Login

Sign Up

Cost Calculator

Selected Printer

Select a printer

Number of Pages

1

Delivery Option

☒ Self Pickup

☐ Class Delivery (+\$10)

Payment Method

redits

Cost Summary

Printing Cost:

\$0.00

Delivery Fee:

\$0.00

Total Cost:

\$0.00

Proceed to Payment

Contact Us:

SmartPrint

Smart Printing for Modern Campuses

Quick Links

Printers

Upload

Calculator

Help & Support

Contact Us

Email: support@smartprint.edu

Phone: (123) 456-7890

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Dashboard:

SmartPrintAdmin

HomePrintersPrint JobsProfileAdminLogin

Admin Dashboard

Manage Users

Manage Printers

Print Jobs

Reports

System Settings

Admin Dashboard

Manage the SmartPrint system and monitor real-time statistics.

Total Users

4

Active Printers

1

Jobs Today

0

Total Revenue

\$0.00

Recent Users

View All

ID	Name	Email	Role	Status	Actions
1	Aadil Mubasshar	aadilmubasshar@gmail.com	USER	Inactive	View
2	test test	test@test.com	ADMIN	Inactive	View
3	auth test	auth@test.com	USER	Inactive	View

Profile Page:

SmartPrint

HomePrint DocumentsPrintersPrint Jobs

test@test.com

T

test test

test@test.com

Edit Profile

Change Password

Profile

My Print Jobs

Payment Methods

Account Settings

Profile Information

Account Balance

\$0.0

Add Funds

Total Print Jobs

1

View All

Payment Method

Change

Recent Activity

View All

Date	Activity	Details	Status
2025-05-13 06:28	Print Job	test stand prototype 10 may.pdf	PENDING

Account Settings

Full Name

test test

Email Address

test@test.com

Your email address cannot be changed.

Phone Number

+1 (555) 123-4567

Notification Preferences

Email Notifications

SMS Notifications